Data Collection

By

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Agenda

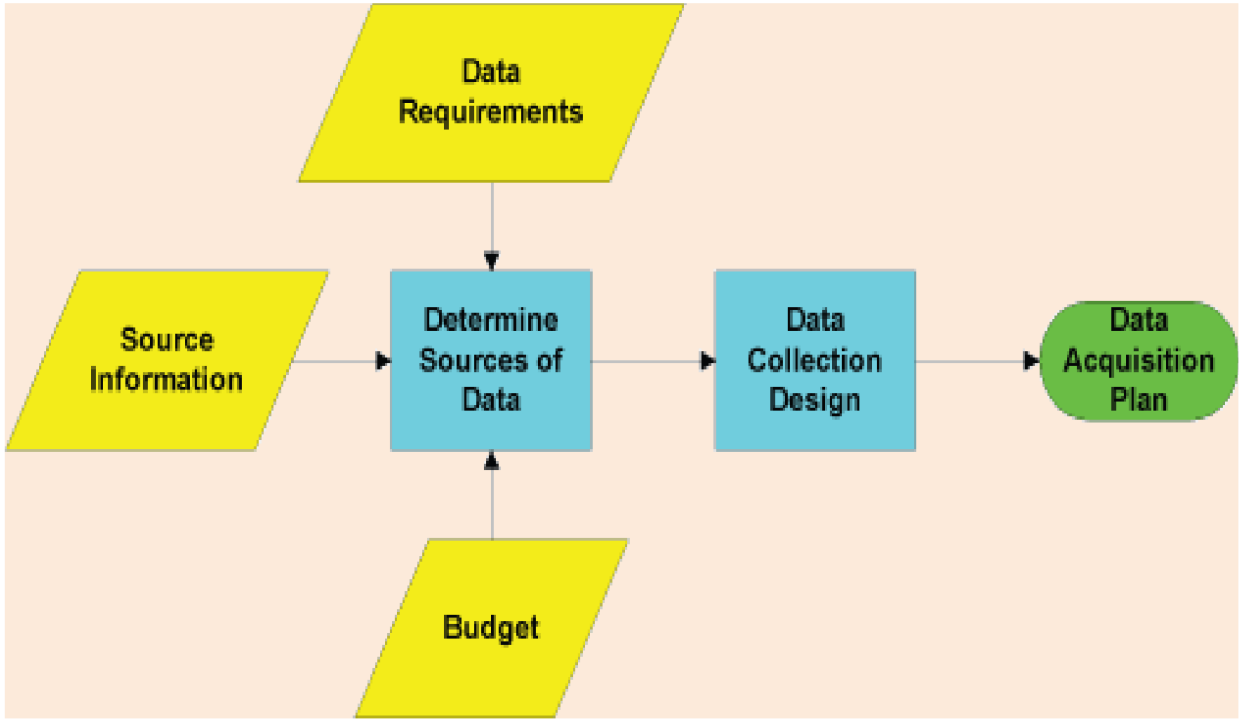
- Primary and Secondary Data
- Data Collection Methods
  - Surveys
  - Experiments
  - Questionnaires
  - Interviews
  - Observation
  - Documents
- Data Preparation
- Data Sampling and Its Techniques
- Evaluation and Validation of Data

- Data is
  - one of the most important and vital aspect of any research studies
  - the basic unit in statistical studies
  - every research is based on data which is analyzed and interpreted to get information.
- Data can be quantitative or qualitative values of a variable
- Data are the basic inputs to any decision making process.
- Data can be numbers, images, words, figures, facts or ideas
- Data collection is a term used to describe a process of preparing and collecting data.
- Purpose of Data Collection
  - To obtain information
  - To keep on record
  - To make decisions about important issues
  - To pass information on to others

Statistics is a tool for converting ***data*** into ***information***

Factors to be Considered Before Collection of Data (plan)

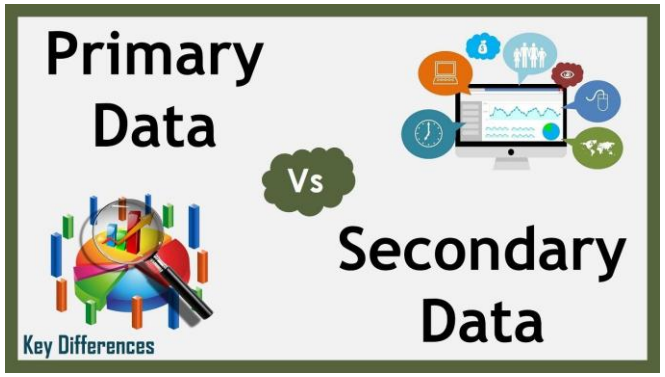
- **Objective** and scope of the enquiry
- **Sources** of information
- **Technique** of da collection
- **Unit** of collection



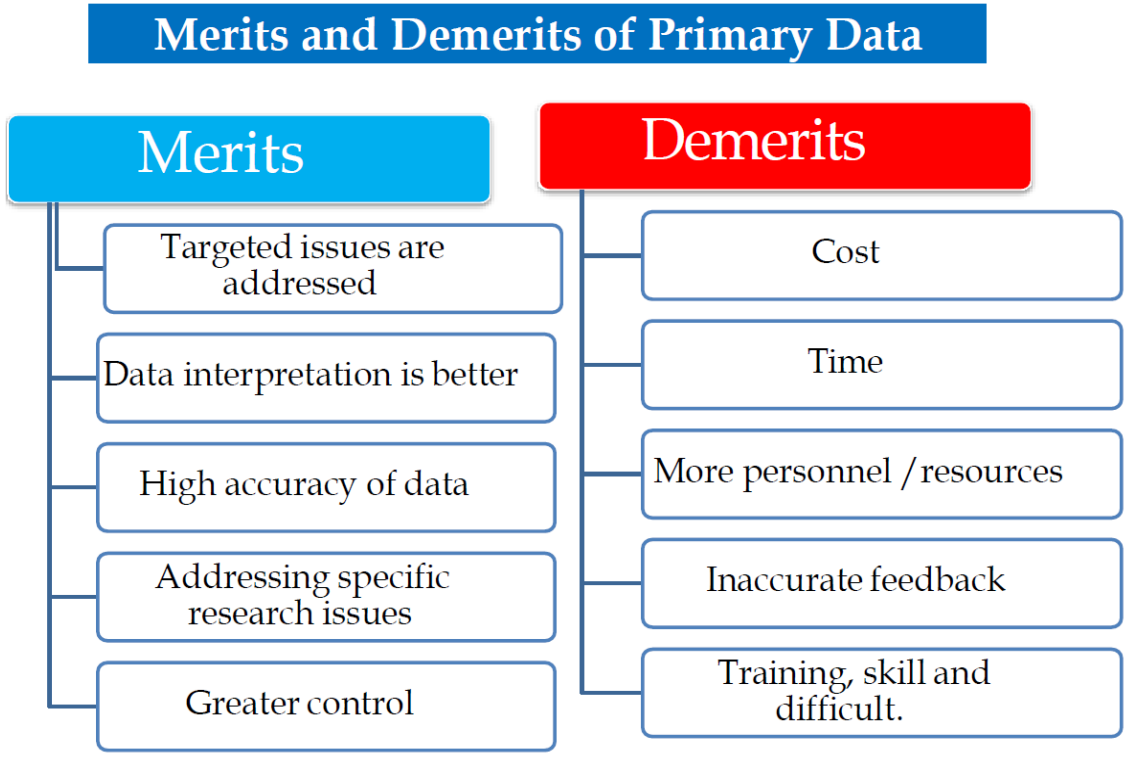
Primary and Secondary Data

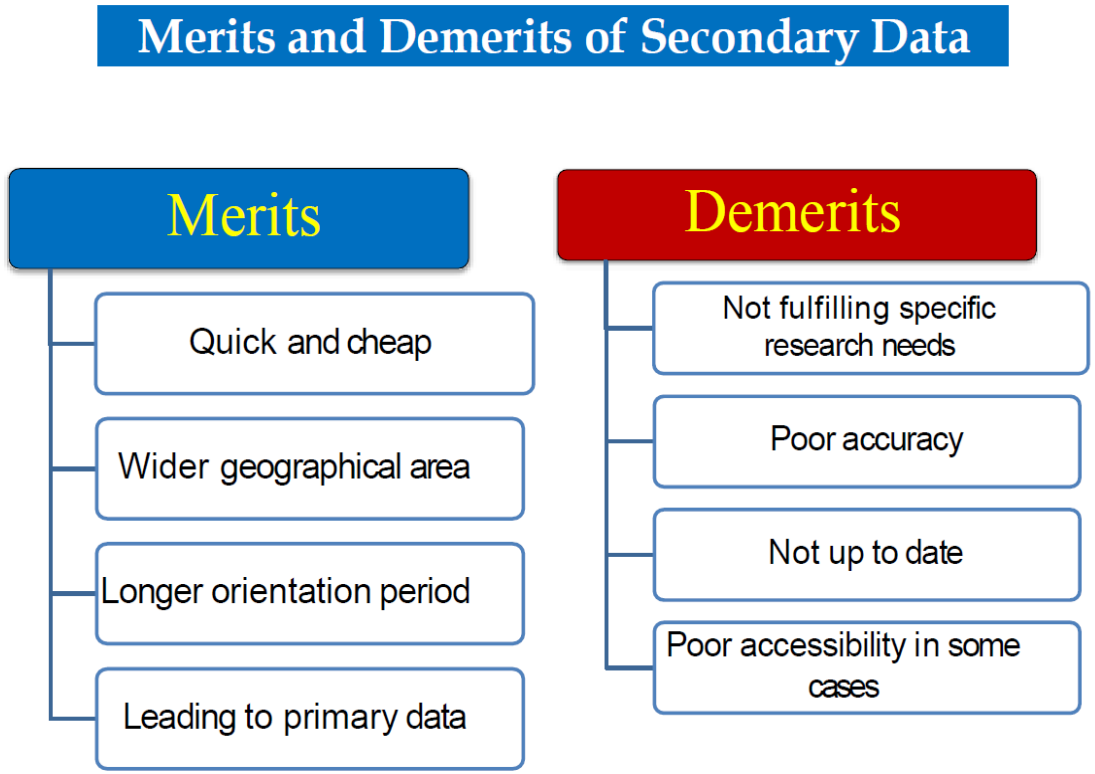
- Primary Data
  - Primary data are original observations collected by the researcher or by his agent for the first time for any investigation and it is collection of data from first hand information.
  - This type of primary data is mostly pure and original data.
  - The primary data is the one type of important data it is useful for current studies as well as for future studies
- **Secondary data** are those which have already been collected by someone else and which have through some statistical analysis

Secondary data *is data that have already been collected for some other purposes, while* ***primary data*** *is data which can be collected specifically for the purpose of answering a research question(s) and to meet research objectives*



Primary vs Secondary Data





Data Collection

Strategies

- No one best way: decision depends on:
  - What you need to know: *numbers or stories*
  - Where the data reside: *environment, files, people*
  - Resources and time available
  - Complexity of the data to be collected
  - Frequency of data collection
  - Intended forms of data analysis

Rules for Collecting Data

- Use multiple data collection methods
- Use available data, but need to know
  - how the measures were defined
  - how the data were collected and cleaned
  - the extent of missing data
  - how accuracy of the data was ensured

Data Collection Tools/Methods: Surveys

- Survey is conducted to obtain data from a large group of people (or events), in a standardized and systematic way.
- Example: Opinion polls survey ~1000 people to predict voting intentions of an entire country.
- Survey is not limited to the use of **questionnaire** for its data generation but also **interviews, observations and documents** can be used.
- An obvious example is the literature survey, which examines printed and online published documents or articles.
- The survey strategy is mostly associated with the **philosophical paradigm of positivism**, since it is looking for patterns and generalizations, but it can also be used with **interpretive and critical research**.
- Widely used in computer engineering research, often featured in empirical studies in computer science journals.
- Surveys can evaluate user feedback on software systems, though poorly designed surveys can undermine results.

Planning and Designing Surveys

- The planning and conducting of surveys can be broken down into six different activities under the following headings:
  - data requirements,
  - data generation method,
  - Sampling frame,
  - sampling technique,
  - response rate and non-responses,
  - sample size.

Data requirements

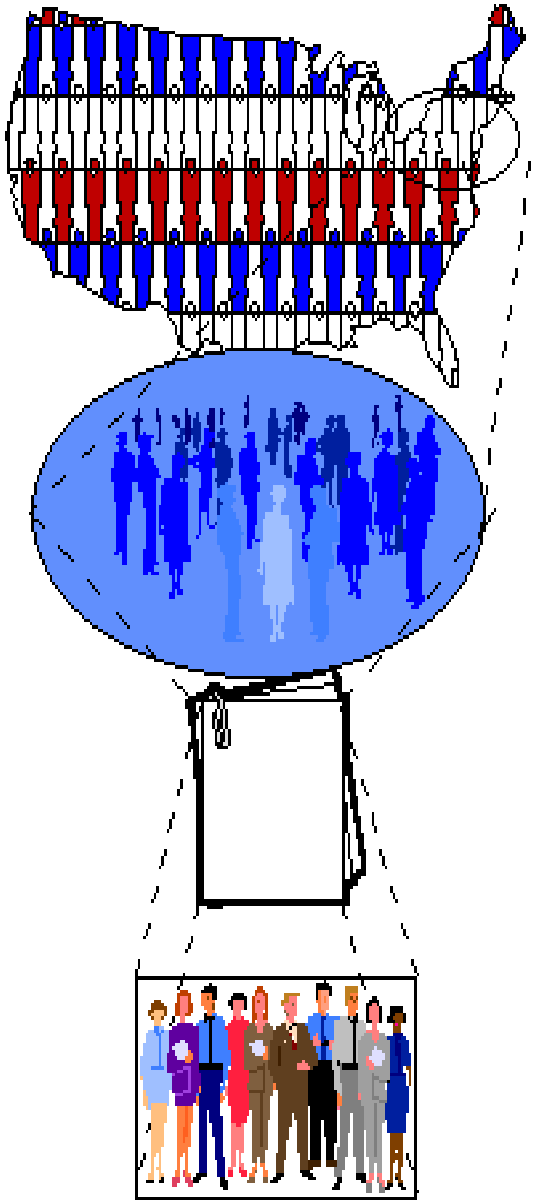
- **Data requirements:** Decide on the type of data to generate, whether directly related to research questions or indirectly (e.g., demographic data like age and gender).
- **Single Opportunity**: In questionnaire- or interview-based surveys, you typically get only one chance to collect data from respondents.
- **Plan Ahead**: Carefully consider all necessary data beforehand, as follow-up opportunities may not be available.
- **Anticipate Analysis:** Think about potential data **analysis methods**, patterns to identify, and interpretations that may arise.
- **Supporting Data**: Ensure you collect additional data that might be needed to support or explain findings during analysis.
- **Thorough planning** prevents missing critical data, ensuring comprehensive and reliable research outcomes.

Data generation method

- **Multiple Data Generation Methods**
  - Surveys are not limited to questionnaires.
  - Other methods include:
    - Interviews
    - Documents
    - Observations
- **Choosing the Right Method**
  - Typically, only **one method** is selected per survey.
  - Decision depends on research objectives, resources, and context.
- **Designing the Data Generation Process**
  - **Questionnaires**: Design the questionnaire carefully.
  - **Interviews**: Plan the interview format and structure.
  - **Observations**: Create detailed observation schedules.
  - **Documents**: Identify existing documents or create new ones; define what to look for.
- **Focus on Precision**
  - Clearly define how the chosen method will generate the required data.
  - Ensure alignment with research goals.

Who do you want to generalize to?

The Theoretical Population



What population can you get access to?

The Study Population

How can you get access to them?

The Sampling Frame

Who is in your study?

The Sample

SAMPLING BREAKDOWN

Sampling frame

- A **list or collection** of the entire population (people, events, or documents) that could be included in your survey.
- Used to select a **sample** for research.
- **Examples of Sampling Frames**
  - IT help-desk logs (e.g., all user queries), telephone directories, attendance registers, electoral rolls (voter registers), membership lists from professional organizations, purchased mailing lists (e.g., "all schools in PU").
- **Importance of a Good Sampling Frame**
  - Must be **up-to-date**, **accurate**, and **complete**.
  - Should cover the **entire population** of interest.
  - Example: A telephone directory is unsuitable for studying homeless populations.
- **Challenges and Biases**
  - **Exclusions**: Some individuals may not be listed (e.g., unlisted phone numbers).
  - **Geographic limitations**: A single town's directory cannot represent a whole country.
  - **Relevance**: Ensure the frame aligns with the research question.
- **Practical Considerations**
  - Choose a sampling frame that matches your research goals.
  - Be aware of potential biases and limitations in the frame.

Sampling Technique

- The method used to select individuals or events from the **sampling frame**.
- **Probability Sampling**:
  - Ensures a **representative sample** of the population.
  - Each member has a known, non-zero chance of being selected.
  - Allows for **generalizations** to the wider population.
  - **Used when representativeness and generalizability are critical.**

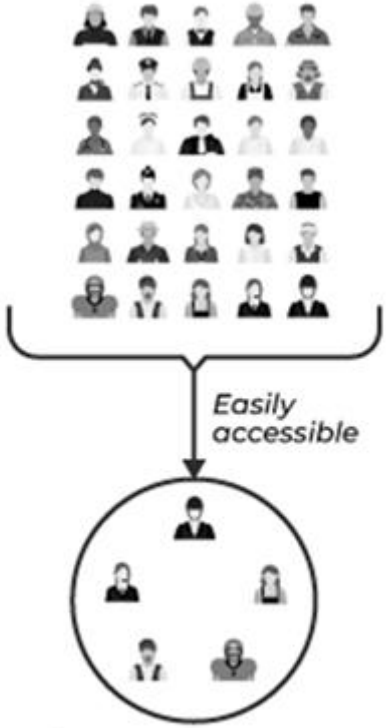
Table 7.1 Sampling techniques

Probabilistic	Non-probabilistic
Random	Purposive
Systematic	Snowball
Stratified	Self-selection
Cluster	Convenience

- **Non-Probability Sampling**:
  - Does not guarantee a representative sample.
  - Members may have unique characteristics not shared by the population.
  - Provides a **weak basis** for generalizations.
  - **Used when feasibility or specificity is more important than generalizability.**
- **Key Considerations**
  - **Research Objectives**: Choose a technique that aligns with your goals.
  - **Resources**: Probability sampling may require more time and effort.
  - **Population Characteristics**: Ensure the technique suits the population being studied.

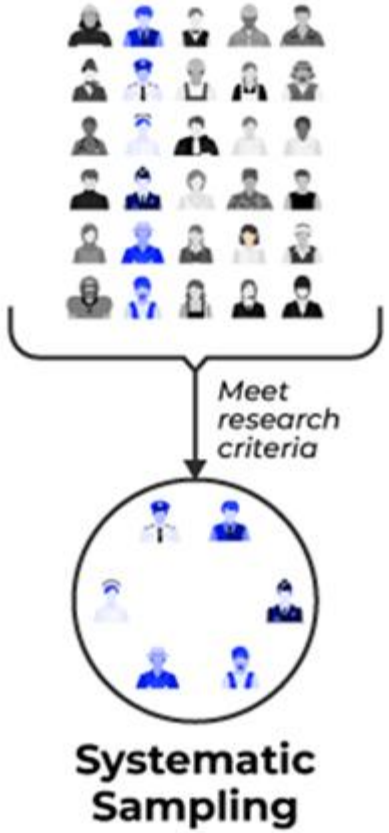
Random Sampling

- Selection of people or events **completely at random**.
- Simple Method for Small Populations
  - Write all names on slips of paper.
  - Place them in a hat and draw the required number of names.
- Method for Larger Populations
  - Use a **list of random numbers** (from mathematical tables, statistics textbooks, or spreadsheet software).
  - Apply each number to identify:
    - **Page number** in the sampling frame (e.g., telephone directory).
    - **Line number** on the page.
  - Select the corresponding entry for inclusion in the sample.
- Practical Tools
  - Random number lists from books or software.
  - Spreadsheet packages for generating random numbers.



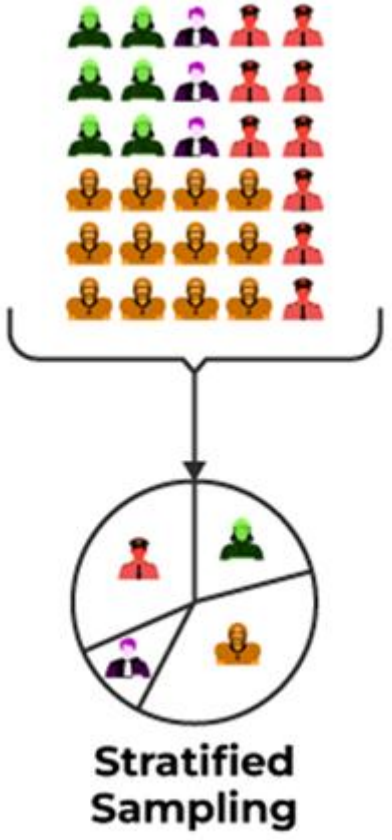
Systematic Sampling

- A sampling technique that builds on **random sampling** by selecting people at **regular intervals** from the sampling frame.
- Process
  - Decide on a **fixed interval** (e.g., every 100th person in a telephone directory).
  - **Randomly select the starting point** (e.g., choose the first name at random).
  - Continue selecting every nth person (e.g., every 100th) from the starting point.
- Combines **randomness** with a **systematic approach** for efficiency.



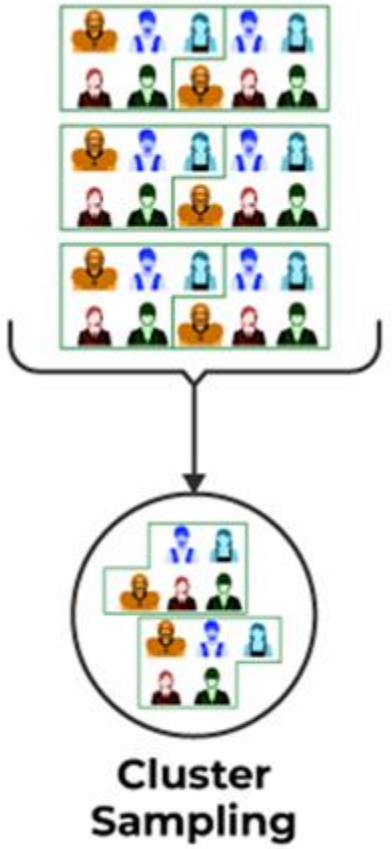
Stratified Sampling

- A sampling technique where the sample is built to **reflect the proportions** of subgroups (strata) in the overall population.
- Process
  - **Divide the sampling frame** into distinct subgroups (e.g., males and females).
  - **Select samples randomly** from each subgroup in proportion to their presence in the population.
- Example
  - **Population**: IT organization employees (70% male, 30% female).
  - **Sample Selection**:
    - 70% of the sample randomly chosen from the male list.
    - 30% of the sample randomly chosen from the female list.
- Ensures the sample **accurately represents** the population's structure.



Cluster Sampling

- A sampling technique where the population is divided into **naturally occurring clusters** (e.g., towns, schools, bars).
- Samples are drawn from selected clusters to reduce costs and effort.
- Process
  - **Identify clusters** (e.g., towns, schools, organizations).
  - **Randomly select clusters** from the population.
  - **Sample within clusters** using other techniques (e.g., random or stratified sampling).
- Example
  - **Research**: National survey on home computer use.
  - **Clusters**: Towns.
  - **Method**:
    - Randomly select several towns.
    - Sample individuals within those towns.
- **Advantages**
  - **Cost-effective**: Reduces travel and logistical expenses.
  - **Efficient**: Focuses on specific areas or groups.
- **Limitations**
  - **Reduced representativeness**: Clusters may not fully reflect the diversity of the entire population.
  - **Trade-off**: Balance between cost savings and sample accuracy.

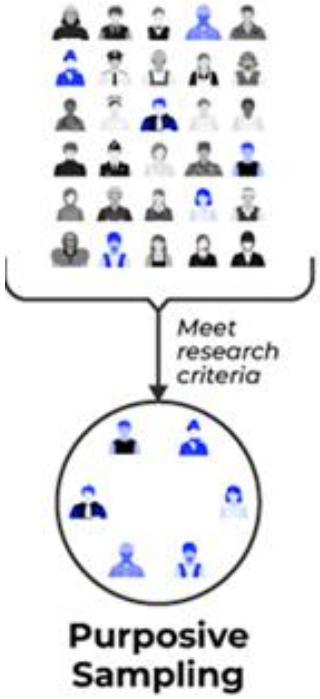


Non-Probability Sampling

- A sampling technique used when a **representative sample** is not feasible or necessary.
- When to Use Non-Probability Sampling
  - **Feasibility**: Time and costs of obtaining a representative sample are too high.
  - **Lack of Information**: Insufficient knowledge about the population (e.g., size, characteristics).
  - **Accessibility**: Difficulty in contacting individuals from a probability sample
  - **Research Focus**: Exploration of a topic in depth, including **extreme, unusual, or atypical cases**.
- Purpose
  - Not aimed at **generalizing** to a larger population.
  - Focuses on **exploring specific issues** or raising diverse perspectives.
- Examples
  - **Interpretive Research**: Deliberately including unique or extreme cases to explore a wide range of issues.
  - **Critical Research**: Examining atypical instances to challenge assumptions or norms.
- Prioritizes **depth** and **specificity** over representativeness.

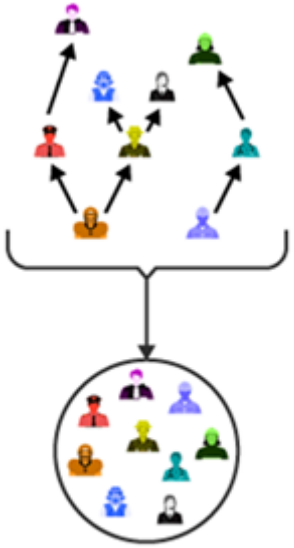
Purposive Sampling

- A **non-probability sampling** technique where the researcher **deliberately selects** instances or individuals likely to provide valuable data.
- Purpose
  - Focuses on **specific characteristics** or **diverse perspectives** rather than representativeness.
  - Aims to explore a wide range of issues, including **extreme or unusual cases**.
- Process
  - Researcher **hand-picks** samples based on the research objectives.
  - Chooses instances that offer **variety** or **depth** of information.
- Example
  - **Research Topic**: Attitudes toward online purchasing.
  - **Sample Selection**:
    - Regularly shop online.
    - Never shopped online.
- Prioritizes **relevance** and **depth** over random selection.



Snowball Sampling

- A **non-probability sampling** technique where the researcher starts with one individual from the target population and expands the sample through referrals.
- Process
  - **Step 1**: Identify and gather data from an initial respondent.
  - **Step 2**: Ask the respondent for referrals to others who fit the research criteria.
  - **Step 3**: Approach the referred individuals and repeat the process.
  - The sample **snowballs** in size as more referrals are added.
- Advantages
  - Useful for accessing **hard-to-reach** or **hidden populations** (e.g., computer hackers).
  - Leverages **trust and introductions** from initial respondents.
  - Can be combined with **purposive sampling** to ensure specific criteria (e.g., gender, employment type) are met.
- Example
  - **Research Topic**: Beliefs and motivations of computer hackers.
  - **Process**:
    - Start with one active hacker willing to participate.
    - Use their referrals to identify and contact other hackers.
- Relies on **networks and connections** within the target population.

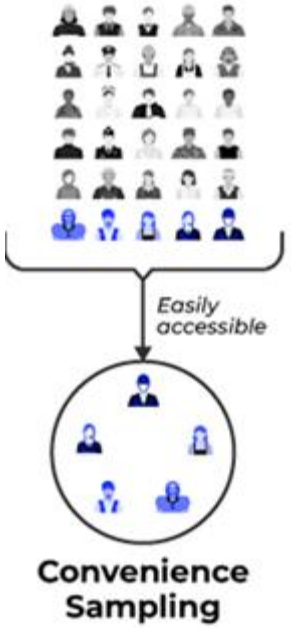


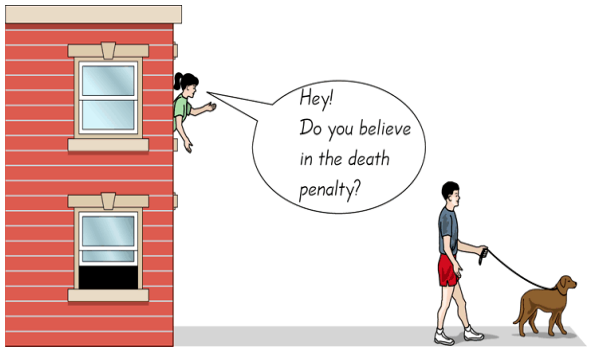
Self-Selection Sampling

- A **non-probability sampling** technique where respondents **volunteer** to participate in the research after seeing an advertisement or call for participants.
- Process
  - Researcher **advertises** the study and its requirements.
  - Individuals who are interested and meet the criteria **self-select** to participate.
- Advantages
  - Useful when the researcher **does not know how to contact potential respondents**.
  - Easy to implement and cost-effective.
- Limitations
  - **Bias**: Participants often have **strong feelings** about the topic, making them unrepresentative of the wider population.
  - **Motivation**: Some volunteers may participate for **personal benefit** (e.g., students seeking higher marks from their tutor).
- Example
  - **Research Topic**: Survey on a specific issue.
  - **Process**:
    - Advertise the study online or through posters.
    - Collect data from individuals who volunteer to participate.
- Relies on **voluntary participation**, which may introduce bias.

Convenience Sampling

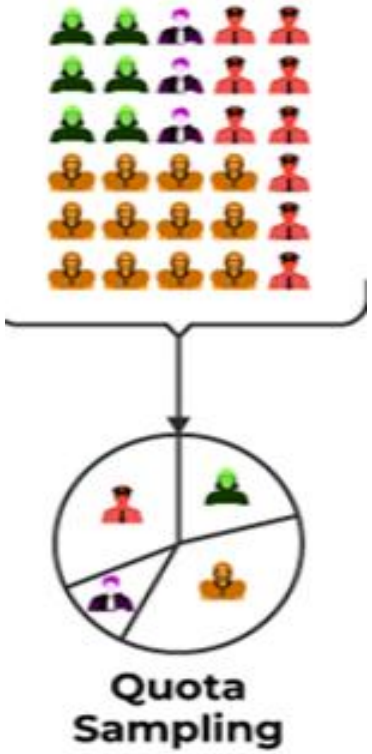
- A **non-probability sampling** technique where researchers select respondents who are **easy to reach** or **willing to participate**.
- Process
  - Researchers choose participants based on **accessibility** and **availability**.
  - Often used when time, cost, or logistical constraints are a concern.
- Advantages
  - **Quick and easy** to implement.
  - **Cost-effective** for preliminary or exploratory research.
- Limitations
  - **Bias**: Participants may not be representative of the wider population.
  - **Criticism**: Using convenience samples (e.g., students) as surrogates for other groups (e.g., professionals) is often seen as poor research practice.
- Prioritizes **ease of access** over representativeness





Quota sampling

- **Quota sampling** is the nonprobability equivalent of stratified sampling.
    - First identify the stratums and their proportions as they are represented in the population
    - Then convenience or judgment sampling is used to select the required number of subjects from each stratum.



Response rate and non-responses

- Response Rate Challenges
  - Low response rates are common (e.g., 10-30%).
  - Strategies are needed to **increase participation**.
- Strategies to Improve Response Rates
  - **Over-sample**: Include more of groups likely to have low response rates.
  - **Clear Explanation**: Explain the purpose and value of the survey to encourage participation.
  - **Personal Delivery**: Hand-deliver questionnaires and explain face-to-face.
  - **Convenience**: Provide stamped, addressed envelopes for postal surveys.
  - **Follow-Up**: Send reminders to non-respondents to politely request participation.
- Analyzing Non-Responses
  - Collect data on **characteristics of non-respondents** (e.g., demographics, behavior).
  - Assess whether non-responses introduce **bias** into the sample.
  - Determine if non-responses are **meaningful** (e.g., reflect disinterest or lack of relevance).
- Importance of Monitoring
  - Track who has and hasn't responded to ensure follow-up actions.
  - Ensure the final sample is as **representative** as possible.

Sample Size

- Importance of Sample Size
  - Determines the **reliability** and **generalizability** of research findings.
  - Must account for **non-response rates** to ensure the final sample is adequate.
- **Rule-of-Thumb** for Small-Scale Research
  - Aim for a **minimum sample size of 30**.
  - Smaller samples (<30) make **statistical analysis unreliable** (e.g., calculating averages or percentages).
- Reporting Results for Small Samples
  - Avoid using **percentages alone** for samples under 30.
  - Always include **actual numbers** (e.g., "3 out of 30" instead of "10%").
- Generalizing to the Population
  - Larger samples are needed for **generalizability**.
  - Sample size depends on:
    - **Population size**.
    - **Desired confidence level** (typically 95%).
    - **Accuracy range** (e.g., ±3%).

Table 7.2 Target population sizes and sample sizes

Target population size	Required sample size for 95 per cent confidence level and +/– 3 per cent accuracy range
50	47
5000	760
100,000	888
900,000	895

Source: www.geneseesurvey.com/web/webcalc.asp

Sample size formulae for reporting precision

For single mean : $n = Z^2_{\alpha} S^2 / d^2$

where S=sd (σ)

For a single proportion : $n = Z^2_{\alpha} P(1-P)/d^2$

Where , $Z\alpha$ =1.96 for 95% confidence level

$Z\alpha$ = 2.58 for 99% confidence level

Problem 1 (Single mean)

A study is to be performed to determine a certain parameter(BMI) in a community. From a previous study a sd of 46 was obtained.

If a sample error of up to 4 is to be accepted. How many subjects should be included in this study at 99% level of confidence?

Answer

$$n = (Z_{\alpha/2})^2 \, \sigma^2 / d^2$$

σ: standard deviation = 46

d: the accuracy of estimate (how close to the true mean)= given sample error =4

$Z_{\alpha/2}$: A Normal deviate reflects the type I error.
For 99% the critical value =2.58

$$n = \frac{2.58^2 \text{ x } 46^2}{4^2} = 880.3 \sim 881$$

Problem 2 (Single proportion)

It was desired to estimate proportion of anemic children in a certain preparatory school. In a similar study at another school a proportion of 30 % was detected.

Compute the minimal sample size required at a confidence limit of 95% and accepting a difference of up to 4% of the true population.

Answer

$n = (Z_{\alpha/2})^2 \, p(1\text{-}p) \, / \, d^2$

p: proportion to be estimated = 30% (0.30)

d: the accuracy of estimate (how close to the true proportion) = 4% (0.04)

$Z_{\alpha/2}$: A Normal deviate reflects the type I error

For 95% the critical value =1.96

$$\mathrm{n} = \frac{1.96^2 \text{ x } 0.3(1-0.3)}{(0.04)^2} = 504.21 \sim 505$$

Evaluating Survey-Based Research

Advantages of Surveys

- **Wide Coverage**
  - Provides **inclusive coverage** of people or events.
  - Results are likely to be **representative** of the wider population.
  - Enables **generalized conclusions**.
- **Efficiency**
  - Produces **large amounts of data** quickly and at a **low cost**.
  - Costs and time can be **predicted in advance**, aiding project planning.
- **Quantitative Data**
  - Suitable for **quantitative analysis**, which is often viewed as credible.
  - Appeals to readers who value numerical data.
- **Replicability**
  - Surveys can be **repeated** with other samples or over time.
  - Supports **scientific validation** of findings through replication.
- **Accessibility**
  - Postal, web, or document-based surveys are ideal for researchers with **limited interpersonal skills**.

Disadvantages of Surveys

- **Lack of Depth**
  - Focuses on **breadth** rather than **depth** of coverage.
  - Limited detail on complex research topics.
- **Overemphasis on Quantifiable Data**
  - Tends to focus on **measurable aspects**, potentially overlooking qualitative insights.
- **Snapshot Nature**
  - Provides **static snapshots** of specific points in time, rather than capturing ongoing processes or changes.
- **No Cause-and-Effect Analysis**
  - Can only show **associations**, not causation (e.g., older managers prefer paper, but age may not be the cause).
- **Limited Observation**
  - In postal, telephone, or Internet surveys, researchers cannot assess **body language** or **response accuracy**.

Experiments

- Investigate **cause-and-effect relationships**.
- Aim to **prove or disprove** a causal link between a factor and an observed outcome.
- Commonly used in **physics, chemistry, and metallurgy**.
- Central to the **scientific method** and **positivist research**.
- Process of Developing a Hypothesis
  - Start with a **theory** about the topic of interest.
  - Formulate a **testable statement** (hypothesis) based on the theory.
  - Hypotheses are of the form: **"Factor A causes B."**
  - **Examples:** The Watson-Klein-Bloggs algorithm enables quicker data processing than the current algorithm.
- Focuses on **empirical testing** to establish **causal relationships**.

Experimental Research

- Researchers make **precise and detailed observations** of outcomes and changes when a factor is introduced or removed.
- Process of Experimentation
  - **Step 1**: Observe or measure a factor.
  - **Step 2**: Manipulate circumstances (introduce or remove a factor).
  - **Step 3**: Re-observe or re-measure the factor to identify changes.
- Focuses on establishing **causal links** between two or more factors.
- **Identification of Causal Factors**
  - Distinguishes between:
    - **Independent Variable**: The cause (factor being manipulated).
    - **Dependent Variable**: The effect (outcome being measured).
  - Goes beyond identifying **correlations** to determine **causation**.
- Explanation and Prediction
  - Uses **theories** to explain causal links.
  - Enables **prediction** of future outcomes if the causal relationship is proven.
- Repetition and Validation
  - Experiments are **repeated** under varying conditions to ensure reliability.
  - Rules out the influence of **external factors** (e.g., faulty equipment).

- Planning and Conducting Experiments

Hypotheses

- Role of Hypotheses in Experiments
  - A hypothesis is a **testable statement** that forms the basis of an experiment.
  - It predicts a **relationship between variables** and is **tested empirically through field data**.
- Characteristics of a Hypothesis
  - Written as a **prediction** about the relationship between factors.
  - Must be **empirically testable** (can be supported or rejected by evidence).
- Examples of Hypotheses
  - **General Prediction**: When factor A occurs, B will happen.
  - **Causal Relationship**: An increase in D causes a decrease in C.
  - **Scientific Example**: Water will freeze when the temperature drops to 32°F (0°C).
- Key Considerations
  - Clearly define the **variables** to be controlled and measured.
  - Ensure the hypothesis is **specific, measurable, and falsifiable**.

Independent and dependent variables

- **Independent Variable**
  - The factor that is **manipulated** by the researcher.
  - Its size, number, or characteristics exist **independently** and are not affected by other variables.
- **Dependent Variable**
  - The outcome that is **measured** or observed.
  - Changes as a result of changes to the independent variable.
- The **independent variable** is manipulated to observe its effect on the **dependent variable(s)**.
- Experiments can involve **multiple independent variables**, but this increases the complexity of statistical analysis.



Controls

- To ensure that **only one factor** (the independent variable) causes the observed change.
- Eliminate or account for other variables that could influence the results.

Methods of Controlling Variables

- **Eliminate the Factor**: Exclude participants or conditions that could skew results. Example: Exclude students with prior programming experience in a study on learning programming skills.
- **Hold the Factor Constant**: Keep certain variables consistent across the experiment. Example: Use participants of the same age if age might affect outcomes.
- **Random Selection**: Randomly select participants to balance out individual differences (e.g., age, gender, beliefs).
- **Control Groups**: Set up two groups:
  - **Control Group**: No manipulation of the independent variable.
  - **Experimental Group**: Independent variable is manipulated.
  - Compare outcomes to isolate the effect of the independent variable.
- **Blind Studies**:
  - Prevent bias by ensuring participants and researchers do not know who is in the control or experimental group.
  - Example: Use a computer program to allocate test materials and analyze results without revealing group assignments.

- **Example of Control in Practice**
  - **Experiment**: Testing a new teaching approach's impact on exam results.
  - **Control**: Ensure the exam and marking criteria remain unchanged to rule out other causes of improved results.
- Controls ensure the **validity** and **reliability** of experimental results by isolating the effect of the independent variable.

Observation and Measurement

- **To measure changes** in dependent variables caused by manipulating the independent variable.
- Essential for identifying cause-and-effect relationships.
- Types
  - Project Data: Person-hours, cost, time to completion.
  - Self-Report Responses: Questionnaires about feelings, ratings, or perceptions.
  - Behavioral Counts: Frequency of specific behaviors (e.g., asking for help, accessing online systems).
  - Technical Metrics: Number of bugs in code, time to process data files.
- Pre-Test and Post-Test Measurements
  - **Before-and-After Comparison**: Measure dependent variables **before** (pre-test) and **after** (post-test) manipulating the independent variable.
  - Example: Measure customer satisfaction **before and after** introducing a new system to detect changes.
- **Quantitative Data**
  - Use **numerical data** to measure change and perform statistical analysis.
  - Example: Rate stress levels on a scale of 1-10 if direct medical measurements (e.g., heart rate) are unavailable.
- **Interrelated Outcome Measures**
  - In IS and computing, outcomes like **productivity, code defects, and delivery times** are often related.
  - Example: Measure **code defect rates** and **analyst stress** alongside delivery times when testing a new development method.
- **Statistical Analysis**
  - Decide **in advance** what to measure and which **statistical tests** to use.
  - Ensures valid and reliable analysis of results.

Internal Validity

- The extent to which observed changes in the dependent variable are **solely due to the manipulation** of the independent variable, not other factors.
- Common Threats to Internal Validity
  - **Selection Bias**: Differences between groups at the start of the experiment may skew results.
  - **History**: External events between pre-test and post-test influence outcomes. Example: Organizational changes (e.g., redundancies) affecting stress levels.
  - **Maturation**: Natural changes in participants over time (e.g., growth, practice, fatigue). Example: Children's improved performance due to age, not experimental manipulation.
  - **Instrumentation**: Faulty or inconsistent measurement tools (e.g., inaccurate devices or interviewer bias).
  - **Experimental Mortality**: Participants drop out, leading to biased samples. Example: Skilled employees leaving a study on technology use.
  - **Reactivity and Experimenter Effects**: Participants change behavior due to awareness of being tested (e.g., seeking training). Experimenter bias (e.g., age, race, or gender influencing results).

External Validity

- The extent to which experimental results can be **generalized** to other situations, populations, or contexts.
- Ensures findings are **not unique** to a specific set of circumstances.
- Allows for **prediction** of similar results in future or different scenarios.
- Achieving High External Validity
  - **Replication**: Repeat the experiment in multiple settings and with different samples.
  - **Careful Design**: Ensure the experiment reflects real-world conditions and uses representative samples.
- Threats to External Validity
  - **Over-Reliance on Special Participants**: Example: Using students as substitutes for managers or the general population. Students may differ in age, education, values, and experiences.
  - **Volunteer Bias**: Volunteers often have unique traits (e.g., higher intelligence, sociability). Results may not apply to non-volunteers.
  - **Small Sample Size**: Too few participants can make it difficult to achieve **statistical significance**.
  - **Non-Representative Participants**: Ensure the sample reflects the population of interest (e.g., use random or stratified sampling).
  - **Non-Representative Test Cases**: Example: Using atypical data files to test a new algorithm may not reflect real-world performance.
- Use **representative samples** and **realistic conditions** to enhance generalizability.
- Avoid over-reliance on **convenient but atypical groups** (e.g., students, volunteers).

Quasi- or Field Experiments

- Experiments conducted in **real-life settings** where researchers cannot fully control variables.
- Focus on **naturally occurring events** rather than laboratory conditions.
- **Purpose is t**o study phenomena in **natural contexts** while maintaining some experimental rigor. Useful for IS researchers studying **social contexts** of system design and use.
- Comparison to True Experiments
  - **True Experiments**: Conducted in controlled laboratory settings. Full control over independent and dependent variables.
  - **Quasi-Experiments**: Conducted in real-world settings. Limited control over variables, making it harder to establish **cause-and-effect relationships**.
- Advantages
  - Provides insights into **real-world scenarios**.
  - Suitable for studying phenomena that cannot be replicated in a lab.
- Limitations
  - **Lack of Control**: Inability to control all variables reduces the ability to establish causation.
  - **Confounding Factors**: Other variables may influence outcomes, making results less conclusive.
- Key Feature
  - Balances **experimental rigor** with **real-world applicability**, though with less control than true experiments.

Uncontrolled Trials

- An **uncontrolled trial** involves introducing a change (e.g., a new system or method) without proper measurement or control of variables.
- It does not meet the criteria for a **true experiment** or a **quasi-experiment**.
- Key Characteristics
  - **No Pre- and Post-Test Measurements**: Fails to measure outcomes before and after the change.
  - **No Control of Variables**: Does not account for other factors that could influence results.
- Limitations
  - **Inability to Establish Causation**: Cannot determine if the change caused the observed outcomes.
  - **Limited Conclusions**: Results are inconclusive and lack scientific rigor.
- Alternative Strategies
  - **Case Study**: In-depth exploration of a specific instance.
  - **Action Research**: Collaborative problem-solving in real-world settings.
- Uncontrolled trials are **not experiments** and provide little meaningful insight.
- Use **controlled experiments**, **case studies**, or **action research** for rigorous and actionable findings.

Evaluating Experiment-based Research

Advantages

- **Scientific Credibility**
  - Widely regarded as the most **scientific** research strategy.
  - Often the only strategy known to those without formal research training.
- **Causal Relationships**
  - The **only strategy** capable of proving **cause-and-effect relationships**.
- **Precision and Control**
  - Laboratory experiments allow for **high precision** in measuring outcomes and analyzing data.
  - Researchers can control variables effectively in a lab setting.
- **Convenience**
  - Researchers can work from their **normal place of work**, avoiding the time and costs of field visits.

Disadvantages

- **Artificial Situations**
  - Laboratory settings may create **unrealistic conditions** that do not reflect real-world scenarios.
- **Difficulty in Controlling Variables**
  - It is often **challenging or impossible** to control all relevant variables, especially in field experiments.
- **Recruitment Challenges**
  - Obtaining a **representative sample** of participants can be difficult.
- **Ethical Concerns**
  - Concealing the research purpose from participants may be necessary to avoid bias, but this is often viewed as **unethical**.

Questionnaires

- **Pre-defined set of questions** (items) arranged in a specific order.
- Respondents provide answers, which are analyzed and interpreted by the researcher.
- Common use in Surveys
  - Often used in **survey research** to collect data from a sample.
  - Sent by post, completed by respondents, and returned for analysis.
- Can be used in other research strategies, such as: **Case studies, Action research, Design and creation**
- Types of Questionnaires
  - **Self-Administered**: Respondents complete the questionnaire independently (e.g., postal or online surveys).
  - **Researcher-Administered**: The researcher asks questions and records responses (e.g., structured interviews). Can be conducted **face-to-face** or **over the telephone**.
    - Researcher's appearance and interaction style may influence responses.
- Provides a **structured and systematic** way to collect data for analysis and generalization.

Why Use Questionnaires?

> **Quality of questionnaires**
>
> Simply stated, the quality of the information obtained from a questionnaire is directly proportional to the quality of the questionnaire, which in turn is directly proportional to the quality of the construction process. (Peterson, 2000, p. 12)

- **Efficiency:** Provide an **efficient way** to collect data from **many people**.
- Best Suited For
  - **Large-Scale Data Collection**: Gathering data from a large number of respondents.
  - **Brief and Uncontroversial Information**: Collecting straightforward, non-sensitive data.
  - **Standardized Data**: Ensuring all respondents answer the same questions with predefined response options.
  - **Literate Respondents**: Respondents must be able to **read and understand** the questions and answers.
  - **Resources Available**: Requires **time and money** for printing, distribution, and collection.
- Common Misconception
  - Novice researchers may think questionnaires are just a **list of questions with multiple-choice answers**.
  - In reality, they require **careful design and construction** to ensure valid, reliable, and useful data.
- Key Considerations for Design
  - **Clarity**: Ensure all respondents interpret questions and answers the same way.
  - **Relevance**: Align each question with **research questions or hypotheses**.
  - **Analysis Plan**: Know in advance how responses will be analyzed.
  - **Structure**: Organize questions in a **logical and coherent manner**.
- Designing a good questionnaire requires **time, effort, and attention to detail** to ensure it meets research objectives and generates high-quality data.

Planning and Designing Questionnaires

- Questionnaires often do not allow for **follow-up questions**, so design must be **right the first time**.
- A **well-designed questionnaire** increases the likelihood of responses. Poorly designed questionnaires may be **discarded** by potential respondents.
- **Key Aspects to Plan and Design**
  - **Question Clarity**: Ensure questions are clear and unambiguous.
  - **Logical Structure**: Organize questions in a coherent and logical order.
  - **Relevance**: Align questions with research objectives.
  - **Ease of Completion**: Make the questionnaire user-friendly to encourage participation.
- Careful planning and design are essential to ensure the questionnaire **collects valid data** and **maximizes response rates**.
- Different aspects of questionnaires are:

Administration for Questionnaires

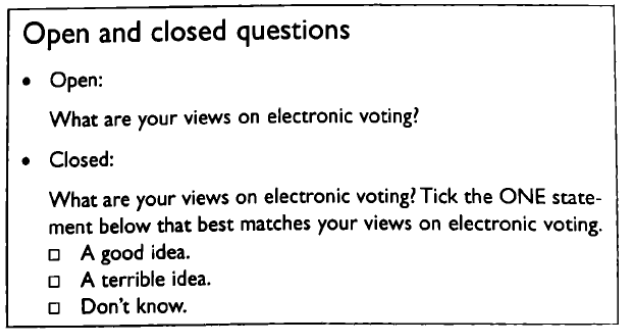
- **Self-Administered Questionnaires**
  - **Advantages**:
    - Saves **time and resources** for the researcher.
    - Allows reaching a **larger number of respondents**.
    - Reduces **bias** from respondents trying to please the researcher.
    - Ensures **consistency** in how questions are presented.
  - **Disadvantages**: Lower **response rates** compared to researcher-administered questionnaires.
- **Researcher-Administered Questionnaires**
  - **Advantages**:
    - Higher **response rates** (harder for respondents to refuse).
    - Allows **follow-up questions** based on previous answers.
  - **Disadvantages**:
    - Risk of **inconsistency** in how questions are asked (e.g., tone, body language).
    - More **time-consuming** and resource-intensive for the researcher.
- Choose the form of administration based on **research goals**, **resources**, and **desired response rates**.
- Ensure **consistency** in question delivery to minimize bias.

Question Content and Wording

- **Content Validity**
  - Ensure questions cover all aspects of the **concepts** you are studying.
  - Example: For **user satisfaction**, include questions on timeliness, accuracy, relevance, ease-of-use, reliability, and user manuals.
- **Preparation**
  - Familiarize yourself with the **research domain** and relevant concepts by reviewing literature.
  - Consider using or adapting questions from **previously used questionnaires** .
  - Avoid **copyright violations** and poorly designed questions.
- **Characteristics of Good Questions**
  - **Brief**: Ideally 20 words or less, while remaining clear.
  - **Relevant**: Each question and word must align with the questionnaire's purpose.
  - **Unambiguous**: Avoid words with multiple meanings or unfamiliar terms.
  - **Specific**: Avoid vague questions or combining multiple questions into one.
  - **Objective**: Avoid leading questions or suggesting a known answer.

- Well-designed questions are **brief, relevant, unambiguous, specific, and objective** to ensure **valid and reliable data**.

Question Types


Open and closed questions

- Open:

  What are your views on electronic voting?

- Closed:

  What are your views on electronic voting? Tick the ONE statement below that best matches your views on electronic voting.

  ☐ A good idea.
  ☐ A terrible idea.
  ☐ Don't know.

- **Types of Data Collected**
  - **Factual Data**: Objective information (e.g., date of birth, job title, number of children).
  - **Opinions**: Subjective views (e.g., thoughts on a computer animation, information system, or societal trends).
- **Open Questions**
  - **Definition**: Allow respondents to answer in their own words.
  - **Advantages**: Capture **rich, detailed responses**. Useful for exploring **unknown or wide-ranging answers**. Easier to prepare than closed questions.
  - **Disadvantages**: Require more effort from respondents. Harder to **code and analyze** due to varied responses.
  - **Design Tip**: Balance space for answers — too little frustrates, too much may discourage responses.
- **Closed Questions**
  - **Definition**: Provide predefined answer options for respondents to choose from.
  - **Advantages**: Easier and faster to **analyze** (pre-coded responses). Save time for respondents.
  - **Disadvantages**: May frustrate respondents if desired answer is not listed. Risk of **superficial responses** without deep thought. May **influence answers** by suggesting options.
  - **Design Tip**: Ensure all possible answers are included to avoid frustration.
- Use **open questions** for depth and exploration, and **closed questions** for efficiency and ease of analysis.

Format of questions and responses

There are many different ways of designing questions and response formats. Here are some of them, with examples.

Yes/no answers

Have you evaluated your company's website in the past 6 months? Yes/No
Does your company use any open source software? Yes/No

Quantity questions

How old were you at your last birthday?
How many children aged 17 or less live in your house?

Agree/disagree with a statement

Do you agree or disagree with the following statement:

The management introduces new computer-based systems here in a way that always takes account of employees' needs. Agree/Disagree.

Degree of agreement or disagreement – the 'Likert scale'

Please tick the ONE box that matches your view most closely:
In my company decision-making is too centralized.

Agree strongly	Agree	Neither agree nor disagree	Disagree	Disagree strongly

Scale questions

For the following statements, please tick the box that matches your view most closely:
Communication between me and my immediate boss is:

Very bad	Bad	About right	Good	Very good

The use of frames in website design is:

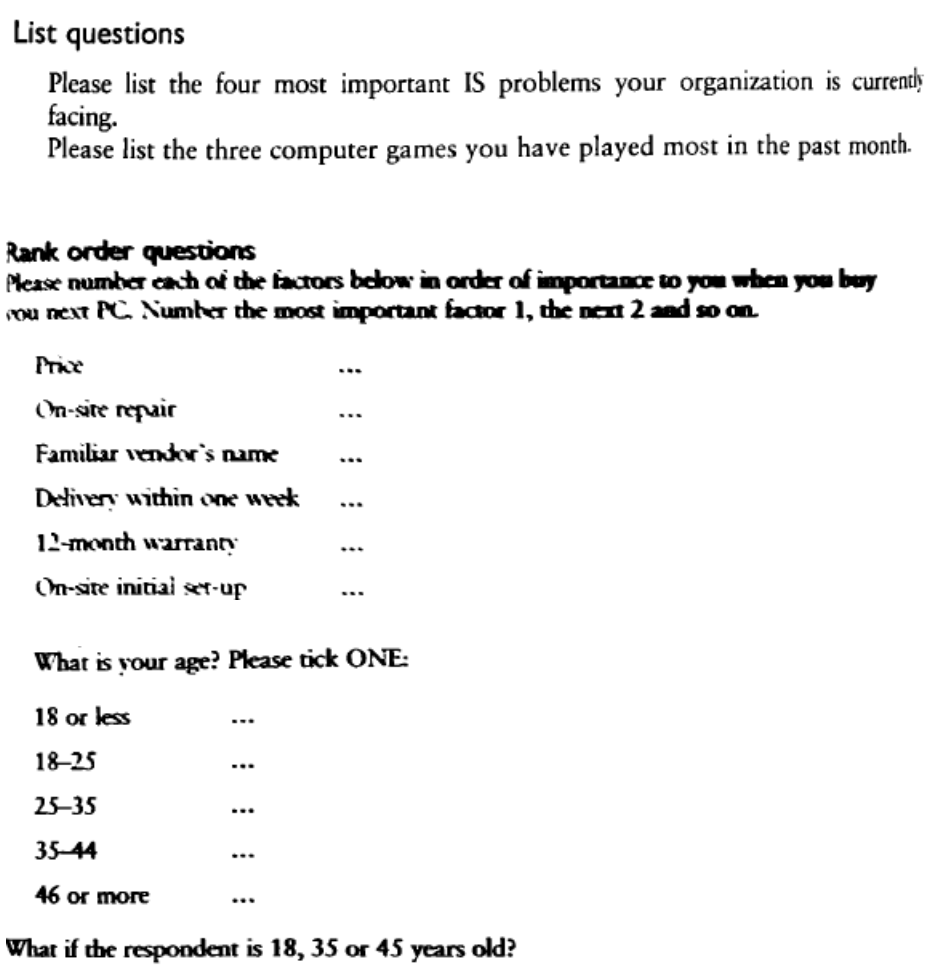
Good practice	Bad practice	Don't know

The 'semantic differential scale'

On each of the lines below, place an X to show how you feel about the lectures on the Knowledge Management Module.

Pace too fast								Pace too slow
Interesting content								Boring content
Good use of slides								Poor use of slides
Too few handouts								Too many handouts

Designing Scales and Question Variety



List questions

Please list the four most important IS problems your organization is currently facing.
Please list the three computer games you have played most in the past month.

Rank order questions
Please number each of the factors below in order of importance to you when you buy you next PC. Number the most important factor 1, the next 2 and so on.

Price	...
On-site repair	...
Familiar vendor's name	...
Delivery within one week	...
12-month warranty	...
On-site initial set-up	...

What is your age? Please tick ONE:

18 or less	...
18–25	...
25–35	...
35–44	...
46 or more	...

What if the respondent is 18, 35 or 45 years old?

- **Avoid Overcomplicating Question Types**
  - Using **too many question types** in one questionnaire can **confuse respondents** and make the design look cluttered.
- **Balancing Variety**
  - A **mix of question types** can keep respondents **engaged and interested**.
- **Designing Scales**
  - For scales (e.g., Likert scales), decide on the **number of points**:
    - Common: **5-point scale**.
    - Range: Up to **9 points**.
  - Include a **neutral option** (e.g., "Don't know" or "Neither agree nor disagree"):
    - **Advantage**: Allows respondents to opt out if unsure.
    - **Disadvantage**: May encourage lazy responses (e.g., ticking "Don't know" without thought).
  - Excluding a neutral option:
    - Forces respondents to take a stance.
    - Risks **alienating** those who genuinely don't know.
- Balance **variety** and **simplicity** in question types.
- Carefully design scales to encourage thoughtful responses while accommodating genuine uncertainty.

Questionnaire Layout and Structure

- **Introduction:**
  - Explain the **purpose** of the questionnaire.
  - Mention the **sponsor** (if applicable).
  - Provide the **return address** and **deadline**.
  - Reassure respondents about **confidentiality** and **voluntary participation**.
  - Thank respondents for their time.
- **Instructions**
  - Include **clear instructions** on how to complete the questionnaire.
  - Provide an **example question and response** for clarity.
- **Question Order**
  - Arrange questions in a **logical sequence** (avoid jumping between topics).
  - Start with **easier and less sensitive questions**.
  - Place **complex or sensitive questions** later.
  - Save **demographic questions** (e.g., age, gender) for the end.
- **Filter Questions**
  - Use filter questions to guide respondents (e.g., "Do you have children? If not, go to Question 9.").
  - Avoid making respondents move **back and forth** excessively.
- **Design and Layout**
  - Ensure the questionnaire is **visually appealing**:
    - Use a **clear layout** with adequate font size and white space.
  - Balance **length**:
    - Not too long to discourage completion.
    - Not too short to miss essential data.
  - If using **optical character recognition (OCR)**, ensure the design meets technical requirements.
- A well-structured and visually appealing questionnaire improves **response rates** and **data quality**

Pre-Test and Pilot for Questionnaires

- **Pre-Test**
  - **Purpose**: Evaluate and refine the questionnaire before full deployment.
  - **Process**: Show the questionnaire to **experts** in the research domain or questionnaire design.
  - **Outcome**: Receive feedback to **improve content, clarity, and structure**.
- **Pilot Test**
  - **Purpose**: Test the questionnaire with a **sample group** to identify issues.
  - **Ideal Scenario**: Use a group **similar to target respondents**.
  - **Practical Scenario**: Often conducted with **colleagues** due to resource constraints.
- **Goals of the Pilot Test**
  - Identify **difficult or confusing questions**.
  - Check for **ambiguity or vagueness** in questions.
  - Ensure **instructions** are clear and easy to follow.
  - Verify that **pre-defined responses** cover all possible answers.
  - Measure the **time taken** to complete the questionnaire.
- **Gathering Feedback**
  - Use **verbal protocol analysis**: Ask respondents to think aloud while completing the questionnaire.
  - Conduct **post-completion interviews** to gather insights.
  - Issue a **follow-up questionnaire** to collect structured feedback.
- Pre-testing and piloting are essential to ensure the questionnaire is **clear, effective, and user-friendly** before full-scale deployment.

Random and Systematic Error

- Random Error
  - fluctuations in the person's current mood.
  - misreading or misunderstanding the questions
  - measurement of the individuals on different days or in different places.
    - These error may cancel out as you collect many samples
- Systematic Error
  - Sources of error including the style of measurement, tendency toward self-promotion, cooperative reporting, and other conceptual variables are being measured.
- So, we have to reduce these errors to prove scientific findings
- How well do our measured variables "capture" the conceptual variables?

Validity and reliability

- **Content Validity**
  - Ensures the questions **cover all aspects** of the domain being studied.
  - Example: For **student satisfaction**, include questions on lectures, seminars, labs, and computer-based support.
  - Use **literature review**, **existing questionnaires**, and **expert panels** to ensure comprehensive coverage.
- **Construct Validity**
  - Ensures the questionnaire **measures what it intends to measure**.
  - Example: A test designed to assess reasoning skills should not simply measure reading speed.
  - **Testing Construct Validity**:
    - Correlate responses with other data (e.g., dissatisfaction with IT department vs. calls to the help desk).
    - Compare results with related measures (e.g., multiple-choice test scores vs. end-of-year exam performance).
- **Reliability**
  - Ensures the questionnaire produces **consistent results** over time or across similar groups.
  - **Challenges**: Respondents may change their views, remember past answers, or deliberately alter responses.
  - **Split-Half Method**: Divide questions into two equivalent groups and compare scores for consistency.
- A good questionnaire must demonstrate **content validity**, **construct validity**, and **reliability** to ensure it measures what it intends to measure and produces consistent results.

The Internet and Questionnaires

- **Global Reach**: Send questionnaires worldwide without postage costs or delivery delays.
- **Flexible Formats**: **Email**: Sent as text or attachments. **Web Forms**: Hosted on websites for user evaluation or research.
- **Design Considerations for Internet Questionnaires**
  - **Shorter Length**:
  - Typing and mouse clicks are slower than writing.
    - Web users prefer quick interactions and may abandon long forms.
    - Costs increase for respondents on pay-as-you-go connections.
    - Risk of losing progress due to network issues.
  - **Save and Resume Feature**: Allow respondents to save progress and submit when ready.
- **Anonymity and Security**
  - **Email Questionnaires**: Reduced anonymity due to visible email addresses. Less suitable for sensitive topics.
  - **Web Forms**: Can be completed anonymously. Risk of multiple submissions by the same respondent. Solutions: Use passwords, but ensure respondents feel anonymous.
- **Response Formats**
  - **Drop-Down Menus vs. Radio Buttons**: Both are commonly used, but no clear preference yet. **Risk with Drop-Downs**: Accidental selection changes using the mouse scroll.
- **Technical Considerations**
  - **Email Attachments**: Assure respondents the attachment is virus-free. Ensure compatibility with common software (e.g., not just Microsoft Word).
  - **Web Forms**: Test on different platforms and browsers to ensure accessibility.
- Internet-based questionnaires offer **global reach** and **flexibility**, but require careful design to address **length**, **anonymity**, and **technical compatibility**.

Questionnaires

Advantages

- **Cost-Effectiveness**
  - Generate **large amounts of data** at relatively low cost.
- **Ease of Use**
  - **Pre-defined answers** (closed questions) make it easy for respondents to complete and researchers to analyze.
- **Flexibility**
  - Can be administered via **post, telephone, email, or web forms**.
- **No Special Skills Required**
  - **Self-administered questionnaires** do not require researchers to have advanced social or communication skills.

Disadvantages

- **Respondent Frustration**
  - **Pre-defined answers** may frustrate respondents, leading to refusal to answer or biased responses.
- **Limited Interaction**
  - Researchers cannot:
    - **Clarify misunderstandings**.
    - **Probe for more detail**.
    - **Verify the truthfulness** of responses.
- **Accessibility Issues**
  - **Self-administered questionnaires** are unsuitable for individuals with:
    - **Poor literacy skills**.
    - **Visual impairments**.

Questionnaires are **cost-effective** and **easy to analyze**, but they may **frustrate respondents** and lack the **flexibility** of interactive methods.

Interviews

- A specific type of conversation with unspoken assumptions.
- **Purpose-driven:** researcher aims to gather information.
- Planned and structured, not random or free-flowing.
- **Key Characteristics of Research Interviews**
  - Researcher controls the agenda and discussion.
  - Interviewee is aware of the research purpose.
  - Responses are typically "**on the record**" unless specified otherwise.
- **Suitable**
  - To obtain detailed or complex information.
  - For open-ended questions requiring flexibility.
  - To explore emotions, experiences, or sensitive issues.
  - When respondents may not share information in written form.
- **Applications**
- Commonly used in case studies and **ethnographies (**qualitative research method focuses on studying cultures, communities, or groups in their natural settings)
  - Can complement surveys (e.g., pre- and post-survey interviews).
  - Useful in design and creation strategies for requirements and feedback.
- **Types**
  - One-to-one interviews.
  - Group interviews (e.g., focus groups).
- Interviews are a versatile data generation method.



Planning and Conducting Interviews

- Interviews are often seen as natural extensions of everyday conversations.
- But research interview require **planning** and a **specific set of skills**.
- Successful research interviews are not intuitive; they demand preparation and skill.

Types of Interviews

Structured Interviews

- Use **pre-determined, standardized questions** for every interviewee.
- Questions are read out exactly as written, with no deviation.
- Responses are often pre-coded or recorded directly.
- Minimal social interaction; avoids indicating personal views.
- Similar to a questionnaire, but conducted verbally.

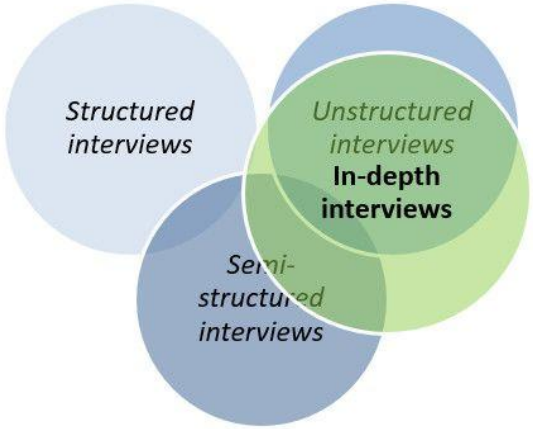
Semi-Structured Interviews

- Based on a **list of themes and prepared questions**.
- Flexibility in question order and follow-up questions.
- Allows interviewees to elaborate and introduce relevant issues.
- Balances structure with conversational flow.

Unstructured Interviews

- Researcher has **minimal control** over the conversation.
- Starts with a broad topic, allowing interviewees to speak freely.
- Focuses on personal accounts, events, behaviors, or beliefs.
- Researcher remains non-intrusive and avoids interruptions.

- **Purpose of Semi-Structured and Unstructured Interviews**
  - Used for **discovery** rather than verification.
  - Ideal for **in-depth investigations** and exploring personal experiences.
  - Not suitable for generalizing findings to a larger population.
- Structured interviews are ideal for standardized data collection.
- Semi-structured and unstructured interviews are better for exploratory research.
- Choice depends on research goals and the need for flexibility.



Researcher's Role and Identity

- Interviewees respond differently based on their perception of the researcher.
- Data generated can be influenced by the researcher's **perceived role and identity**.
- **Factors Influencing Interviewee Responses**
  - Researcher's **profession** (e.g., student, journalist, police detective).
  - **Demographics**: sex, age, ethnic origin, accent, and status.
- **Researcher's Approach**
  - Aim to be **professional, polite, punctual, receptive, and neutral**.
  - Limited control over inherent aspects of identity that may influence responses.
- **Key Considerations for Researchers**
  - **Age gap**: How might it affect the interview dynamics?
  - **Gender or ethnic differences**: Could they lead to reluctance in responding?
  - **Social status or education differences**: How might they impact the interview?
  - **Appearance**: Adjust to fit the context of the interviewees (e.g., formal vs. creative settings).
- Consider adjusting self-presentation to minimize bias and improve data quality.

Interview Preparation

- Plan the **issues to raise** and **questions to ask**.
- Gather **background information** on interviewees and their context.
- **Background Research**
  - For organizational interviews: Review company reports, news articles, and relevant materials.
  - For individual interviews: Learn about their job title, department, or role.
- **Benefits of Background Research**
  - Highlights potential issues to discuss during the interview.
  - Establishes **credibility** as a professional, encouraging interviewees to open up.
  - Helps assess the accuracy of information provided by interviewees.
- **Sharing Topics/Questions in Advance**
  - Sending a list of topics or questions beforehand:
    - Gives interviewees time to prepare their thoughts.
    - Enhances your credibility as a serious researcher.



- Conduct at least one **practice interview** with a friend.
- Thorough preparation ensures a professional and effective interview process.
- Background research and practice interviews are essential for success.

Scheduling

- Clearly explain the **purpose** and **duration** of the interview to the interviewee.
- **Time Management**
  - Avoid underestimating the time required.
  - It's better to finish early than to overrun into the interviewee's next activit
- Interviews are tiring and require write-ups afterward.
- Schedule **no more than three interviews per day**.
- **Interviewee's Time Commitment**
  - Limit interviews to **two hours maximum**.
  - For longer discussions, arrange **multiple shorter sessions** to avoid fatigue or frustration.
- Select a location where the interviewee feels **comfortable** (e.g., their office, canteen, or home).
- Ensure the venue is **private, quiet, and free from disturbances**.
- Respect the interviewee's time and energy.

> **Don't take up too much of your interviewee's time!**
>
> 'I have answered three questions, and that is enough,'
>
> Said his father, 'don't give yourself airs!
>
> Do you think I can listen all day to such stuff?
>
> Be off, or I'll kick you downstairs!'
>
> (Lewis Carroll, 'You are old, Father William', *Alice's Adventures in Wonderland* [London: Penguin, 1994; first published 1865], Chapter 5)

Recording Interviews

- Relying on memory alone is **unreliable** due to bias and error.
- Use recording methods to ensure accuracy and completeness.
- **Recording Methods**
  - **Field notes**: Minimum requirement, capturing key points and context.
  - **Audio tape recording**: Provides a complete record of verbal responses.
  - **Video recording**: Captures non-verbal communication and context.
- **Field Notes**
  - Include:
    - Interviewee's statements.
    - Context (location, atmosphere).
    - Observations of non-verbal communication (e.g., body language).
    - Researcher's thoughts on intent behind statements.
  - Write additional notes immediately after the interview while memory is fresh.
- **Challenges with Field Notes**
  - Difficult to write, think, and ask questions simultaneously.
  - Notes may not capture everything; practice is essential.

- **Audio Tape Recording**
  - **Advantages**:
    - Complete record of verbal responses.
    - Allows researchers to focus on the interview process.
    - Enables other researchers to review or analyze the data.
  - **Disadvantages**:
    - Interviewees may feel nervous or inhibited.
    - Does not capture non-verbal communication or context.
- **Video Tape Recording**
  - **Advantages**:
    - Captures both verbal and non-verbal communication.
    - Provides a more complete record of the interview.
  - **Disadvantages**:
    - Equipment can be expensive and bulky.
    - Often intrusive and may inhibit interviewees.

- Respect interviewee preferences regarding recording methods.
- Use a combination of methods (e.g., audio recording + field notes) for a comprehensive record.

Seating and Equipment Setup

- **Seating Arrangement**
  - Aim for **comfortable interaction**:
    - Sit at a **90-degree angle** to allow eye contact without confrontation.
  - Respect the interviewee's office setup if changes are not possible.
- Set up equipment **quickly and efficiently** to avoid wasting interview time.
- **Preparation of Equipment**
  - Thoroughly check equipment **before the interview**:
  - For audio recording: Use tapes long enough to cover the entire interview without interruption.
- **Professional Etiquette**
  - **Switch off your mobile phone** at the start of the interview.
  - Avoid distractions and maintain focus on the interviewee.
- Proper seating and equipment setup contribute to a smooth and professional interview process.
- Preparation and attention to detail ensure minimal disruptions and a comfortable environment.

The interview

- **Introduction Phase**
  - Begin with **introductions** and clarify the **purpose** of the interview.
  - Obtain **permission to record** and assure **confidentiality** and **anonymity**.
  - Use **small talk** to ease tension (e.g., journey, weather, holiday plans).
- **Starting the Interview**
  - Begin with an **easy question** to build rapport.
  - Gradually move to more complex or sensitive topics later in the interview.
- **Question Design**
  - Use **clear, concise, and jargon-free questions**.
  - Prefer **open-ended questions** (e.g., starting with *what*, *how*, or *why*).
  - Use **closed questions** sparingly for specific facts or confirmations.
- **Aids for Discussion**
  - Use **visual aids** (e.g., photos, screenshots, websites) to stimulate discussion.
  - Describe **scenarios** or ask about **recent events** relevant to the research.

A well-structured and respectful approach ensures a productive and comfortable interview experience.

- **Researcher's Behavior**
  - Show **active listening** through eye contact and leaning forward slightly.
  - Use **silence** effectively to encourage interviewees to elaborate.
  - Employ **prompts**, **probes**, and **checks** to guide the conversation.
- **Maintaining Neutrality**
  - Be **non-judgemental** and avoid showing personal reactions.
  - Use a **neutral tone** and avoid leading questions.
  - Monitor **body language** to avoid conveying bias.
- **Handling Sensitive Situations**
  - Respect the interviewee's right to skip questions or stop the interview.
  - Back off if the interviewee seems uncomfortable or stressed.
- **Closing the Interview**
  - Invite the interviewee to raise **unaddressed points**.
  - **Thank** the interviewee for their time.
  - Offer to send a **summary or transcript** for verification

The interview

Prompts, probes and checks

Examples of prompts:

- Remain silent.
- Repeat your question.
- Repeat the last few words said by your interviewee.

Examples of probes:

- 'Could you give me some examples of that?'
- 'Would an example of that be when …?'
- 'Can you give me more a bit more detail about what you mean by …?'

Examples of checks:

- 'So, let me see if I've understood you correctly …'
- 'What this means, then, is that …'
- 'If I can summarize what I think you've said …'

Transcribing Interviews

- **Purpose of Transcription**
  - Converts audio recordings into **written form** for easier analysis.
  - Allows for **searching** and **organizing** data effectively.
- **Time Commitment**
  - Transcription is time-consuming: assume **5 hours per hour of tape**.
  - Longer if the tape quality is poor or if the transcriber is inexperienced.
- **Benefits of Transcription**
  - Brings the interview **back to life** and aids in **initial analysis**.
  - Provides an opportunity to add **informal notes** (e.g., gestures, atmosphere, thoughts).
- **Accuracy in Transcription**
  - Capture the interviewee's words **as spoken**, including pauses and false starts.
  - Edit slightly for clarity (e.g., remove "erms" and "uhs") when using quotations in reports.
  - Avoid distorting the **meaning** of the interviewee's words.





- **Capturing Non-Verbal Cues**
  - Most transcripts focus on **words only**, losing intonation and non-verbal sounds.
  - Use established conventions to include **richness of speech** if relevant to the research.
- **Organizing Transcripts**
  - Label each transcript with:
    - Interviewee's **name**.
    - **Date** and **location** of the interview.
  - Number each line for easy reference and analysis.
- Transcription is a **laborious but essential** step in qualitative research.
- Ensures accurate and organized data for analysis and reporting.

Checking and Analyzing Interview Data

- Take **notes or transcripts** back to the interviewee for **verification**.
- Ensures **accuracy** of facts and confirms the intended meaning of responses.
- **Cross-Checking Data**
  - Compare interview data with **other sources** (e.g., documents, observations).
  - Helps confirm statements or identify inconsistencies in responses.
- **Identifying Themes**
  - Look for **recurring themes** across multiple interviews.
  - Avoid basing findings solely on one individual's views.
- **Preparing for Data Analysis**
  - After planning, conducting, and transcribing interviews, move to **data analysis**.
  - Analysis can be:
    - **Quantitative**: Counting occurrences of specific keywords.
    - **Qualitative**: Exploring themes and patterns in responses.
    - **Mixed**: Combining both quantitative and qualitative approaches.
- Verification and cross-checking ensure **reliability** and **validity** of interview data.
- Identifying themes and patterns strengthens the analysis and findings.

Group Interviews

- Involves interviewing **3 to 6 people together**.
- Focuses on **group interaction** and discussion, not individual questioning.
- **Advantages**
  - Generates **consensus views** and **new insights**.
  - Encourages **varied responses** as participants challenge or inspire each other.
  - Useful for **brainstorming themes** for further research (e.g., surveys).
- **Disadvantages**
  - Risk of **dominance** by certain members (e.g., men or higher-status individuals).
  - Quieter members may struggle to **express their views**.
  - Participants may express only **socially acceptable opinions**.
- **Selecting Group Members**
  - Choose participants of **similar status** to encourage open discussion.
  - Arrange seating so everyone is **visible** and the researcher is **not the focal point**.
- **Facilitating Group Interviews**
  - Requires **facilitation skills** to ensure everyone is heard.
  - Use **audio recording** to capture discussions, but note that transcription is more complex.
- Group interviews can provide **rich, interactive data** but require careful planning and facilitation.
- Balancing participation and managing group dynamics are critical for success.

Tactical questions for reducing dominance and involving others

- 'Jean, what do you think?'
- 'What do the rest of you think about this?'
- 'How does Mike's point relate to the one you raised, Sarah?'
- 'What do people think of Atta's suggestion?'



Evaluating Interview-based Research

- **Advantages**
  - **Depth and Detail**: Effective for exploring topics in depth.
  - **Low Equipment Needs**: Requires minimal equipment and leverages existing social skills.
  - **Verification**: Researcher can confirm the interviewee's suitability to answer questions.
  - **Flexibility**: Allows adjustment of questions based on the interview flow.
  - **Interviewee Engagement**: Interviewees often enjoy sharing their ideas with a non-critical listener.
  - **Preference Over Questionnaires**: Some respondents find talking easier than writing.
- **Disadvantages**
  - **Time-Consuming**: Includes interview time, transcription, and analysis of unstructured data.
  - **Lack of Reliability**: Researcher influence and context can reduce consistency and objectivity.
  - **Potential Misleading Data**: Focuses on what interviewees say, not necessarily what they do or think.
  - **Artificial Setting**: Interviewees may alter responses due to awareness of being recorded.
  - Require **strong social skills** and **tact** to avoid stress or discomfort for both parties.
  - Not suitable for **generalizing to a large population** due to time and effort constraints.

Observations

- Involves **watching** and **paying attention** to behaviors, interactions, and environments.
- Uses multiple senses: **sight, hearing, smell, touch, and taste**.
- **Purpose**
  - Captures **what people actually do**, not just what they report.
  - Can study **behavior in natural settings** (e.g., meetings, production lines, IT help desks).
  - Also applicable to **inanimate objects** (e.g., software programs, devices).

Types

- **Overt Observation**:
  - Participants know they are being observed.
  - **Advantages**: Ethical, allows consent, researchers can ask questions.
  - **Disadvantages**: Risk of the **Hawthorne Effect** (behavior modification due to observation).
- **Covert Observation**:
  - Participants are unaware they are being observed.
  - **Advantages**: Natural behavior, undisturbed setting.
  - **Disadvantages**: Ethical concerns, risk of exposure, potential danger.
- **Ethical Considerations**
  - **Overt observation** is generally more ethical and safer.
  - **Covert observation** may be acceptable in public spaces if no harm is caused.
- **Approaches to Observation**
  - **Systematic Observation**: Structured and focused on specific behaviors or events.
  - **Participant Observation**: Researcher actively engages in the setting while observing.

- Observation is a **versatile method** for capturing real-world behaviors and interactions.
- Choice between **overt** and **covert** observation depends on research objectives and ethical considerations

Highly systematic observations of pre-defined types of events	↔	Observations of anything and everything
Narrow concentration on particular type of event	↔	Broad focus
Observer takes no part in the proceedings	↔	Observer participates fully in the proceedings
Fact of observations taking place is known to all	↔	Fact of observations taking place is known to none except the researcher
No explanation, or false explanation, given for presence of observer-researcher	↔	Full explanation given for the presence of the observer–researcher
Short duration – could be as little as 5 minutes	↔	Long duration – possibly years
Record-keeping uses only simple note taking	↔	Record-keeping uses technology (e.g. audio tape, camera, stop watch, two-way mirror, computer program)
No feedback given afterwards to the observed	↔	Full feedback given afterwards to the observed

Figure 14.1 Different kinds of observation

Planning and Conducting Systematic Observation

- Involves **pre-defined criteria** for observing specific events or behaviors.
- Uses a **structured schedule** to record frequency, duration, or timing of events
- **Purpose**
  - Generates **quantitative data** through counting or timing.
  - Focuses on **specific behaviors** or events of interest.
- **Examples**
  - **Group Meetings**: Counting the number and type of contributions by each member.
  - **IT Help-Desk Queues**: Timing arrival, wait time, and query resolution for each student.
  - **Individual Behavior**: Observing one person's actions for a set time, then switching to another.
- **Observation Schedule Design**
  - **Frequency of Events**: Counting how often specific behaviors occur in a given time period.
  - **Timing of Events**: Recording the duration of specific events (e.g., program processing time).
  - **Events at Specific Times**: Logging everything happening at predefined intervals (e.g., every 15 minutes).

Date: Observer:

Lab number	Time	No. of students	No. of working computers	No. of out-of-order computers

Figure 14.2 Observation schedule for monitoring computer laboratories

- Systematic observation is a **structured and objective method** for collecting quantitative data.
- Ideal for studying **specific behaviors** or events in a controlled and repeatable manner.

Choosing or Designing an Observation Schedule

- **Using Existing Schedules**
  - Check **literature** or **online resources** for pre-designed schedules.
  - Example: Pre-defined schedules for observing meeting contributions (e.g., new suggestions, peace-making, criticism).
  - Explore **psychology** or **engineering literature** for technical observation schedules.
- **Designing Your Own Schedule**
  - Spend time **observing the situation** to identify relevant events or activities.
  - Ensure the schedule is:
    - **Easily defined**: Clear and obvious behaviors to observe.
    - **Relevant**: Aligned with research objectives.
    - **Comprehensive**: Includes all possible categories.
    - **Unambiguous**: Categories should not overlap.
    - **Easy to record**: Focus on isolated events rather than simultaneous ones.
- Conduct a **pilot study** to test the effectiveness of the observation schedule.
- **Including Contextual Information**
  - Add space to note **contextual factors** that may influence observations:
    - **Seating arrangements** (e.g., who sits where).
    - **Time of day** (e.g., late afternoon fatigue).
    - **Environmental factors** (e.g., room temperature, noise, weather).

- A well-designed observation schedule ensures **accurate, relevant, and efficient data collection**.
- Pilot testing and contextual notes enhance the **reliability** and **depth** of observations.

Date: Time: Observer:

Contextual factors:

Names of group members: Seating arrangements:

A:

B:

C:

D:

E:

Instructions: Note each time a group member does one of the activities

	A	B	C	D	E
Leading, e.g. proposing plans, seeking suggestions					
Constructive, e.g. making helpful suggestions, trying to solve problems					
Drawing in, e.g. encouraging others, supporting others, inviting others to contribute					
Obstructive, e.g. criticizing, disparaging, blocking contributions					
Clarifying/summarizing, e.g. rewording ideas/objectives, checking progress and time remaining					
Joking, e.g. being humorous					

Figure 14.3 Schedule for observing group meetings

Working with other observers

- Systematic observation can be conducted by **trained assistants**, not just the researcher.
- **Training Observers**
  - Ensure assistants understand:
    - The **observation schedule**.
    - The **categories** to observe.
    - The **recording methods**.
  - Conduct a **pilot run** to test the process and ensure clarity.
- **Inter-Observer Reliability**
  - If observers record different results, potential issues include:
    - **Inadequate training**.
    - **Unsuitable observers**.
    - **Poorly designed schedule**.
    - **Inappropriate research method** for systematic observation.
- **Self-Observation**
  - Participants can record their own activities (e.g., daily tasks).
  - **Challenges**:
    - Risk of **inaccurate or dishonest reporting**.
    - Potential for a **distorted picture** of activities.

- Working with other observers increases **efficiency** but requires **proper training** and **schedule design**.
- Ensure **consistency** and **reliability** through pilot testing and clear guidelines.

Sampling in Observational Research

- Observing people or events = conducting a survey via observations.
- Requires careful selection of sample type.
- **Time-Based Sampling**
  - Researchers often use set time periods (e.g., 1-hour blocks).
  - Ensures structured and manageable data collection.
- **Choosing Appropriate Time Blocks**
  - Select time periods that reflect typical behavior.
  - Avoid biased or unrepresentative timeframes (e.g., Saturday evenings for library usage).
- **Avoiding Bias**
  - Ensure observations are representative of the entire population or phenomenon.
  - Example: Observing library usage only on weekends = biased results.

- Sampling strategy must ensure representativeness and avoid bias for valid results.

Sampling, Analysis and Interpretation of Observational Data

- Sampling as a Survey Method
  - Observing people or events = conducting a survey via observations.
  - Requires careful selection of the type of sample to ensure representativeness.
- Time-Based Sampling and Avoiding Bias
  - Researchers often use set time periods (e.g., 1-hour blocks).
  - Choose appropriate time blocks to avoid bias (e.g., observing library usage only on Saturday evening = non-representative results).
- Analysis and Interpretation of Observational Data
  - **Analyzing and Interpreting Data**
    - Analyze observational data and interpret results with clear reasoning.
    - Consider alternative explanations for observed behaviors or events.
  - **Limitations and Further Research**
    - Observations reveal **what** occurred, not **why** it occurred.
    - If multiple plausible explanations exist, additional research (e.g., interviews, experiments) may be needed to determine the most plausible cause.

Participant Observation

- **Overt vs. Covert Observation**
  - **Overt**: People know you are conducting research.
  - **Covert**: People believe you are a "normal" participant, unaware of your research role.
- **Rich Description and Immersion**
  - No pre-defined observation schedule; focus on detailed, unstructured notes.
  - Researcher uses their senses (seeing, hearing, etc.) to experience and reflect on the setting.
- **Connection to Ethnography**
  - Participant observation is often linked to ethnography.
  - Aims to understand **life from within the setting, uncovering meanings behind actions and behaviors.**

Participant Observation

Types of participation

- **Complete Observer**
  - Present in the setting (overtly or covertly).
  - Observes without participating (e.g., sitting in a classroom without engaging in activities).
- **Complete Participant**
  - Covert observation; becomes a member of the group.
  - Requires necessary credentials (e.g., age, skills, qualifications) to blend in.
- **Participant-Observer**
  - Shadows individuals, observing and occasionally participating (e.g., helping with tasks).
  - Gains trust to understand behaviors and feelings.
- **Practitioner-Researcher**
  - Combines job role with research.
  - Benefits: No need to negotiate access; already familiar with the setting.
  - Challenges: Requires permission, awareness of biases, and balancing job and research roles.

Process of Participant Observation

- **Time-Consuming Nature**
  - Longer time in the setting = deeper understanding.
  - Start with non-selective observation to grasp the overall context.
- **Focused Observation and Theory Building**
  - Identify significant or unusual patterns as they emerge.
  - Develop a tentative theoretical model of the situation.
- **Testing and Revising Theories**
  - Look for unexpected or contradictory events to refine your theory.
  - Focus on issues participants find important to understand their perspectives.
- **Purpose of Participant Observation**
  - Aim to develop a theory about the situation, not just narrate personal experiences.

Field Notes in Participant Observation

- Essential for recording observations, as memory alone is insufficient.
- Notes should be made as soon as possible after observing relevant events.
- **Challenges for Covert Observers**
  - Covert observers may need creative solutions (e.g., using trips to the toilet) to record notes privately.
- **Content of Field Notes**
  - Record observations, thoughts on the research process, and emerging analysis.
  - Include reflections on the researcher’s role, feelings, and potential impact on the situation.

Rules of the Game in Observational Research

- **Informed Consent and Awareness**
  - Participants may not fully understand that ***everything*** **they** do or say might be recorded.
  - Clarify that observation includes both formal and informal settings (e.g., meetings and casual conversations).
- **Ethical Dilemmas**
  - Even if participants say, **"this is not to be used**," the information cannot be erased from memory.
  - Researchers must weigh the ethical implications of reminding participants about observation, as it may alter their natural behavior.
- **Transparency and Reminders**
  - Clearly explain the scope of observation when obtaining permission.
  - Gently remind participants of your presence if they become too comfortable, but balance this with the need to observe natural behavior.

Dangers and difficulties

- Participant observation and covert research involve significant risks, including
  - physical danger,
  - emotional stress,
  - loss of objectivity, and
  - legal exposure.
- Proper support systems and self-awareness are essential to navigate these challenges effectively.

Validity

Challenges to Validity

- Single researcher reliance: Readers may question if findings are unbiased and replicable.
- Human limitations:
  - **Selective recall**: Remembering some details while forgetting others.
  - **Selective perception**: Noticing certain things while ignoring others.
  - **Accentuated perception**: Heightened sensitivity to specific details based on personal experiences or current state.
- **Inevitable bias**: Being selective and partisan is inherent in observational research.

Strengthening Validity

- **Verbatim Quotations**:
  - **Use direct quotes** from participants instead of summarizing.
- **Triangulation**:
  - Observe **multiple people** performing the same action to check consistency.
  - **Combine methods** (e.g., interviews) to confirm observational findings.
- **Reflexivity**:
  - Continuously reflect on your role, assumptions, and impact on the setting.
  - Include reflections in the research write-up to help readers assess replicability.

Evaluating Observation-Based Research

Advantages of Systematic Observation

- **Accuracy**:
  - Reveals what people actually do, not just what they say they do.
- **Efficiency**:
  - Collects large amounts of quantitative data quickly.
  - Data is pre-coded, ready for analysis.
- **Delegation and Scalability**:
  - After training, schedules can be used by others, enabling simultaneous observations in multiple locations.
- **Insights into Mundane Activities**:
  - Captures data on behaviors participants may overlook (e.g., time spent deleting spam emails).

Disadvantages of Systematic Observation

- **Limited Scope**:
  - Focuses only on overt behavior, ignoring intentions, meanings, or reasons.
- **Oversimplification**:
  - Assumes behavior can be easily categorized, potentially missing complexity.

Evaluating Observation-Based Research

Advantages of Participant Observation

- **Cost-Effective**:
  - Requires little equipment, making it inexpensive.
- **Rich Insights**:
  - Provides deep, holistic understanding of complex social settings.
- **Understanding Beliefs and Intentions**:
  - Reveals participants' perspectives and motivations.
- **Convenience for Practitioner-Researchers**:
  - Allows research to be conducted alongside normal job duties.

Disadvantages of Participant Observation

- **Presence Dependency**:
  - The researcher must be present during events; anything missed cannot be recorded.
- **Access Barriers**:
  - Some settings require specific credentials, limiting the researcher's ability to observe.
- **Risks to the Researcher**:
  - Exposure to physical, social, legal, or psychological dangers.
- **Reliability Issues**:
  - Findings depend heavily on the researcher's perspective, making replication difficult.
- **Limited Generalizability**:
  - Observations may be unique to the specific setting, reducing applicability to other contexts.

Evaluating Observation-Based Research

Ethical Concerns

- **Feedback Difficulties**:
  - Hard to provide feedback to participants, raising ethical questions about using people for research without benefiting them.
- **Exploitation Concerns**:
  - Participants may feel used for the researcher’s gain, with no direct benefit to themselves.

Documents

- Documents can be treated as another source of data, **an alternative to interviews, observations and questionnaires**.
- We can divide them into two types:
  - **Found Documents**:
    - Pre-existing materials (e.g., production schedules, profit/loss accounts, internal directories, job descriptions, procedure manuals).
  - **Researcher-Generated Documents**:
    - Created for research (e.g., field notes, photographs, models, diagrams, logs).
- **Organizational Documents**
  - **Formal Records**: Personnel records, sales figures, shareholder reports, meeting minutes.
  - **Informal Communications**: Notes, memos, emails.
  - **Public Records**: Electoral registers, birth/marriage/death records.
- **Personal Documents**
  - Diaries, logs, letters, phone texts, emails.
  - Everyday items: bus tickets, shopping lists, till receipts, credit card bills.
  - Unique sources: gravestone inscriptions, graffiti, tee-shirt slogans.
- **Publications**
  - **Academic Literature**: Books, journal articles, conference papers.
  - **Popular Literature**: Newspaper articles, magazine pieces, brochures.
  - **Guides**: Programmer manuals, software guides.
- **Secondary Data**
  - Archived research data and field notes.
  - Government surveys (e.g., censuses, unemployment counts, labor market surveys).
  - Internal organizational research (e.g., job-satisfaction surveys, suggestion schemes).
- **Multimedia Documents**
  - **Visual Sources**: Pictures, photographs, diagrams, animations, comic strips, signposts, videos, models.
  - **Aural Sources**: Sounds, music.
  - **Electronic Sources**: Screenshots, websites, computer games, online community archives.

Planning and Conducting Document-based Research

- Documents can be used in various research strategies:
  - Design and creation, surveys, case studies, ethnographies, experiments, and action research.
- Documents can verify or challenge data from other methods (e.g., interviews, observations).
- Example: Personnel reports can correct an interviewee's estimate of employee numbers.
- Surveys of documents are a valid research method.
  - Examples: Literature reviews or analyses of specific types of websites.

Obtaining Access to Documents

- **Many documents** (e.g., government reports, company annual reports, public archives) are easily accessible via libraries or the web.
  - No need for consent, appointments, or ethical considerations for public domain data.
  - Some research archives may require a fee (e.g., market research reports).
- **Challenges with Sensitive or Confidential Documents**
  - Internal documents (e.g., company memos, police records) require negotiation with data holders and individuals described in the data.
  - Researchers must assure confidentiality and ethical use of the data.
  - Data protection laws may restrict access to individual files without explicit consent.
- **Access to Secret Documents**
  - Highly confidential documents (e.g., strategic plans, hidden financial accounts) are rarely accessible to researchers.
  - Covert participant observation may be the only way to access such data.
- **Copyright Issues for Found Data**
  - Written permission is required to reproduce copyrighted materials (e.g., images, music).
  - Fees may apply, but academic research purposes can sometimes reduce or waive costs.
  - If copyright holders cannot be traced, include a disclaimer in the research.
- **Ethical and Legal Considerations**
  - Researchers must respect confidentiality, avoid harm, and comply with data protection laws.
  - Transparency and ethical practices are crucial when accessing and using sensitive or copyrighted materials.

Using secondary data and research data archives

- Tradition of Reusing Data
  - Government and publicly funded statistical reports provide reusable raw data and analyzed findings.
  - Museums, art galleries, and libraries archive images and documents for research use.
  - Research data archives catalog and preserve data from previous studies, often accessible online for free.
- Quantitative vs. Qualitative Data Archives
  - **Quantitative Data:** Widely archived (e.g., survey and questionnaire data).
  - **Qualitative Data**: Historically under-archived, with 90% at risk of being lost (e.g., stored in researchers' homes or offices).
- Uses of Secondary Data
  - **Reusing Existing Data**: Saves time by using pre-collected survey data or interview transcripts.
  - **Follow-Up Studies**: Enables re-contact with interviewees for comparative analysis.
  - **Re-Analysis**: Apply new questions, statistical techniques, or theoretical frameworks to uncover new patterns or themes.
- Methodological Insights: Learn from researchers' field notes to refine or adapt methodologies.
- Verification and Critique: Examine original data to validate or challenge earlier findings, identifying potential biases or assumptions.

Authenticity and Trustworthiness

- Consider when and how the document was created:
  - Immediately after an event or based on indirect information?
- Evaluate the purpose of the document:
  - Was it created to influence shareholders, managers, or another audience?
- Relevance to Research
  - Determine if the document aligns with your research goals.
  - Remember that documents provide a partial view, tailored for a specific audience and purpose.
  - Example: Meeting minutes often reflect consensus, not the debate or dissent behind decisions.
- Limitations of Personal Accounts
  - Diaries, field notes, photographs, and videos are subjective and may **not represent objective reality**.
  - Digital media can be **easily altered**, **raising questions about authenticity**.
- Critical Evaluation of Statistics
  - **Counting Challenges**: Some data (e.g., population counts) may be incomplete due to unregistered or hard-to-count groups.
  - **Vested Interests**: Organizations may manipulate statistics (e.g., sales figures, help-desk metrics) to present favorable outcomes.
  - **Judgement-Based Data**: Statistics often rely on individual decisions (e.g., cause of death coding), reducing reliability.

Analyzing Documents

- Two Broad Approaches
  - Documents as Vessels:
    - Focus on the content (data) within documents.
    - Analyze with skepticism about reliability and author bias.
  - Documents as Objects:
    - Treat documents as entities, examining their production, consumption, and circulation.
- Content Analysis (Quantitative): Count occurrences of specific elements (e.g., warnings about cost overruns in project reports).
- Thematic Analysis (Qualitative): Identify and explore themes within a selection of documents.
- Semiotic Analysis (For Images): Study the symbolism and meaning embedded in visual content.
- Analyze the lifecycle of documents:
  - Who produces, consumes, and circulates them?

Evaluating Document-based Research

- Advantages
  - **Cost and Time Efficiency**: Data can be obtained quickly, cheaply, and conveniently, often without appointments or permissions.
  - **Unobtrusive Collection**: Many documents are pre-existing and readily available, reducing ethical concerns.
  - **Longitudinal Studies**: Regularly published reports enable long-term analysis without extensive time commitments.
  - **Permanence and Scrutiny**: Public domain documents allow other researchers to verify findings, enhancing credibility.
- Disadvantages
  - **Evaluation Challenges**: Requires careful assessment of the author, source, purpose, and production process.
  - **Lack of Objectivity**: Documents may not provide an objective view, despite their official appearance.
  - **Access Issues**: Some documents are expensive, inaccessible, or temporary (e.g., internet-based content).
  - **Misalignment with Research Goals**: Secondary data may not align with your research questions.
  - **Inconsistent Categorization**: Changes in survey methods (e.g., unemployment definitions) can hinder longitudinal comparisons.

Why Data Preprocessing?

- Data in the real world is dirty
  - incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
  - noisy: containing errors or outliers
  - inconsistent: containing discrepancies in codes or names
- No quality data, no quality mining results!
  - Quality decisions must be based on quality data
  - Data warehouse needs consistent integration of quality data
- A well-accepted multidimensional view: Accuracy, Completeness, Consistency, Timeliness, Believability, Value added, Interpretability, and Accessibility
- Broad categories:
  - intrinsic, contextual, representational, and accessibility.

Major Tasks in Data Preprocessing

- Data cleaning
  - Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies
- Data integration
  - Integration of multiple databases, data cubes, or files
- Data transformation
  - Normalization and aggregation
- Data reduction
  - Obtains reduced representation in volume but produces the same or similar analytical results
- Data discretization
  - Part of data reduction but with particular importance, especially for numerical data

- Data Cleaning
  - Data cleaning tasks
    - Fill in missing values
    - Identify outliers and smooth out noisy data
    - Correct inconsistent data